Ankur Aditya

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RESEARCH INTERESTS

Multimedia Systems, Computer Vision, Computer Networks, RF/Wireless Sensing

EDUCATION

University of Massachusetts (UMass), Amherst **Sept '22 - Present**
Doctor of Philosophy (MS/PhD), Computer Science CGPA: 3.7/4
Relevant Courseworks: Distributed and Operating Systems, Intelligent Visual Computing, Advanced Algorithms, Neural Networks, Advanced Information Assurance

Indian Institute of Technology Kanpur (IIT-K), Kanpur **Jan '21 - May '22**
Master of Science (MS), Electrical Engineering CGPA: 8.67/10
Transferred to UMass Amherst for continuation of Ph.D.
Relevant Courseworks: MIMO Wireless Communications, Detection and Estimation Theory, Error Correcting Codes, Representation and Analysis of Random Signals, Machine Learning for Wireless Communications

International Institute of Information Technology (IIIT), Bhubaneswar **Aug '16 - June '20**
Bachelor of Technology (B.Tech), Electronics and Telecommunication Engineering CGPA: 9.06/10
Relevant Courseworks: Antenna and Wave Propagation, Mobile Communications.

HONORS AND AWARDS

- Recipient of an ACM Travel Grant for The 22nd ACM Workshop on Hot Topics in Networks (HotNets'23).
- Awarded a government-based merit scholarship for two consecutive years (2016-2020) during undergraduate studies.
- Ranked 1st in the Electronics and Telecommunication department during undergraduate studies, placing in the top 5% of the graduating class.

PUBLICATIONS

- X. Chen, **Ankur Aditya**, Z. Lei, D. Ganesan. 2023. Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate. In The 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23), November 12–17, 2023, Istanbul, Turkiye. ACM, New York, NY, USA, 13 pages.
- **Ankur Aditya** et al., D2RF: Beamforming with Flexible Antenna on a Textile Substrate (**Under Review**).

RESEARCH EXPERIENCE

Research Assistant, Computer Science, University of Massachusetts **May '25 - Present**
Advisor: Prof. Prashant Shenoy *Amherst, MA, USA*

- **Project: VoluStream: Reliable Volumetric Video Streaming System for Videoconferencing**
  - Designed a hybrid loss-recovery pipeline combining selective TCP retransmissions for critical frames with a novel neural recovery module for non-critical frames, enabling reliability under bursty packet loss.
  - Built a codec-agnostic neural loss-recovery module, implemented using a Video Vision Transformer (ViViT) pipeline that reconstructs corrupted frames from compressed streams, enforces temporal consistency, and integrates seamlessly with modern video codecs without requiring encoder modifications.
  - Implemented and evaluated VoluStream within aiortc, a Python implementation of WebRTC.
  - Achieved real-time frame recovery on a desktop-grade GPU with negligible PSNR/SSIM degradation, outperforming conventional error concealment and retransmission-only baselines.

Research Assistant, Computer Science, University of Massachusetts **Jan '24 - Aug '24**
Advisor: Prof. Deepak Ganesan *Amherst, MA, USA*

- **Project: D2RF: Beamforming with Flexible Antenna on a Textile Substrate (Under Review)**
  - Developed a novel predictive model using a Vision Transformer (ViT) to estimate the electromagnetic (EM) characteristics of flexible antennas.
  - Demonstrated the successful application of computer vision methods to a real-world physics problem, achieving comparable accuracy to EM simulators (e.g., HFSS) but orders of magnitude faster.
  - Built a comprehensive simulation library (D2EM) mapping physical deformations to EM properties, enabling a 2x improvement in physical beamforming and over 20x in large-scale simulations.

Research Assistant, Computer Science, University of Massachusetts **Jan '23 - May '23**
Advisor: Prof. Deepak Ganesan *Amherst, MA, USA*

- **Project: Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate**
  - Explored solutions for signal degradation caused by physical deformations (curvature, bending) in flexible antenna arrays mounted on fabric.
  - Developed CurtainNet, an end to end system delivering over 12dB of beamforming gain and extending wireless coverage radius by up to 20m by correcting for these deformations.

Research Intern, Electrical Engineering, Indian Institute of Technology **Dec '19 - June '20**
Advisor: Prof. K. V. Srivastava *Kanpur, UP, India*

- **Project: Design of Microwave Absorbers with Transmission Band using Frequency Selective Surfaces**
  - Designed and simulated a novel, ultra-thin (5.57mm) metamaterial absorber with an integrated transmission band, achieving a peak insertion loss of only 0.167dB.
  - Implemented an Absorption-Transmission-Absorption (A-T-A) rasorber structure, realizing two broadband absorption bands (5.27 to 11.63 GHz, 15.35 to 17.11 GHz) and a narrow, highly selective transmission band (11.63 - 15.35 GHz).
  - Verified performance and robustness through full-wave High Frequency Structure Simulator (HFSS) simulations.

TEACHING EXPERIENCE

Teaching Assistant, Computer Science, University of Massachusetts **Sep '22 - Present**
Amherst, MA, USA

- Reasoning Under Uncertainty (Fall '22, Fall '23, Spring '25 and Fall '25)
- Introduction to Computation (Fall '24)
- Scalable Web Systems (Summer '23)
- Introduction To Data Science (Summer '23)

ACADEMIC SERVICES

- **Shadow Reviewer**, ACM MobiCom 2024
- **Volunteer**, Graduate Admission Program Committee, Indian Institute of Technology Kanpur, India

SKILLS

- **Programming Languages:** ◇ Python ◇ MATLAB
- **Frameworks & Libraries:** ◇ PyTorch ◇ NumPy ◇ Open3D ◇ scikit-learn ◇ TorchCodec
- **Antenna Simulation Tools:** ◇ Ansys HFSS ◇ CST Microwave Studio
- **Other Tools:** ◇ Docker ◇ Git ◇ GNU Radio ◇ Blender ◇ PyRealSense ◇ MeshLab ◇ LaTeX